

38) Five friends have access to a chat room. Is it possible to determine who is chatting if the following information is known? Either Kevin or Heather, or both, are chatting. Either Randy or Vijay, but not both, are chatting. If Abby is chatting, so is Randy. Vijay and Kevin are either both chatting or neither is. If Heather is chatting, then so are Abby and Kevin. Explain your reasoning.

Soln: p : Kevin is chatting q : Heather is chatting r : Randy is chatting
 s : Vijay is chatting t : Abby is chatting

- ① $(p \vee q)$ is True ② $rs' + r's$ is True ③ $t \Rightarrow r$ is True
 ④ $ps + p's$ is True ⑤ $q \Rightarrow (p \wedge t)$ is True

Let t be True. By defn. of conditional, and from ③, r is True.
 By defn. of negation, $\neg r$ is False. By defn. of AND, tr is True.
 Since $(p \wedge r) \vee (\neg p \wedge r)$ is True, by defn. of OR, $p \wedge r$ must be True.
 By defn. of AND, p is True (consistent) and r is True. By defn. of negation, s is False. By defn. of AND, $(p \wedge s)$ is False. Since $(p \wedge s) \vee (\neg p \wedge s)$ is True, by defn. of OR, $\neg p \wedge s$ is True. By defn. of AND, $\neg p$ is True (consistent) and s is True. By defn. of negation, p is False. Since $(p \vee q)$ is True, by defn. of OR, q is True.

$q \rightarrow \text{True}$ $p \rightarrow \text{False}$ $t \rightarrow \text{True}$. By defn. of ~~neg~~ AND, $p \wedge t$ is False.
 By defn. of conditional, $q \Rightarrow (p \wedge t)$ is False. This contradicts ⑤.
 $\therefore t$ is False. By defn. of AND, $p \wedge t$ is False. Since ⑤ is True, by defn. of conditional, q is False. Since ① is True, by defn. of OR, p is True.

By defn. of Negation, $\neg p$ is False. By defn. of AND, $\neg p \wedge r$ is False.
 Since ④ is True, by defn. of OR, $p \wedge s$ is True. By defn. of AND, p is True, and s is True. By defn. of Negation $\neg s$ is False. By defn. of AND, $(p \wedge s)$ is True.
 By defn. of AND, $\neg p$ is True and s is True (consistent). By defn. of negation, r is False.

Ans: Only Kevin and Vijay are chatting